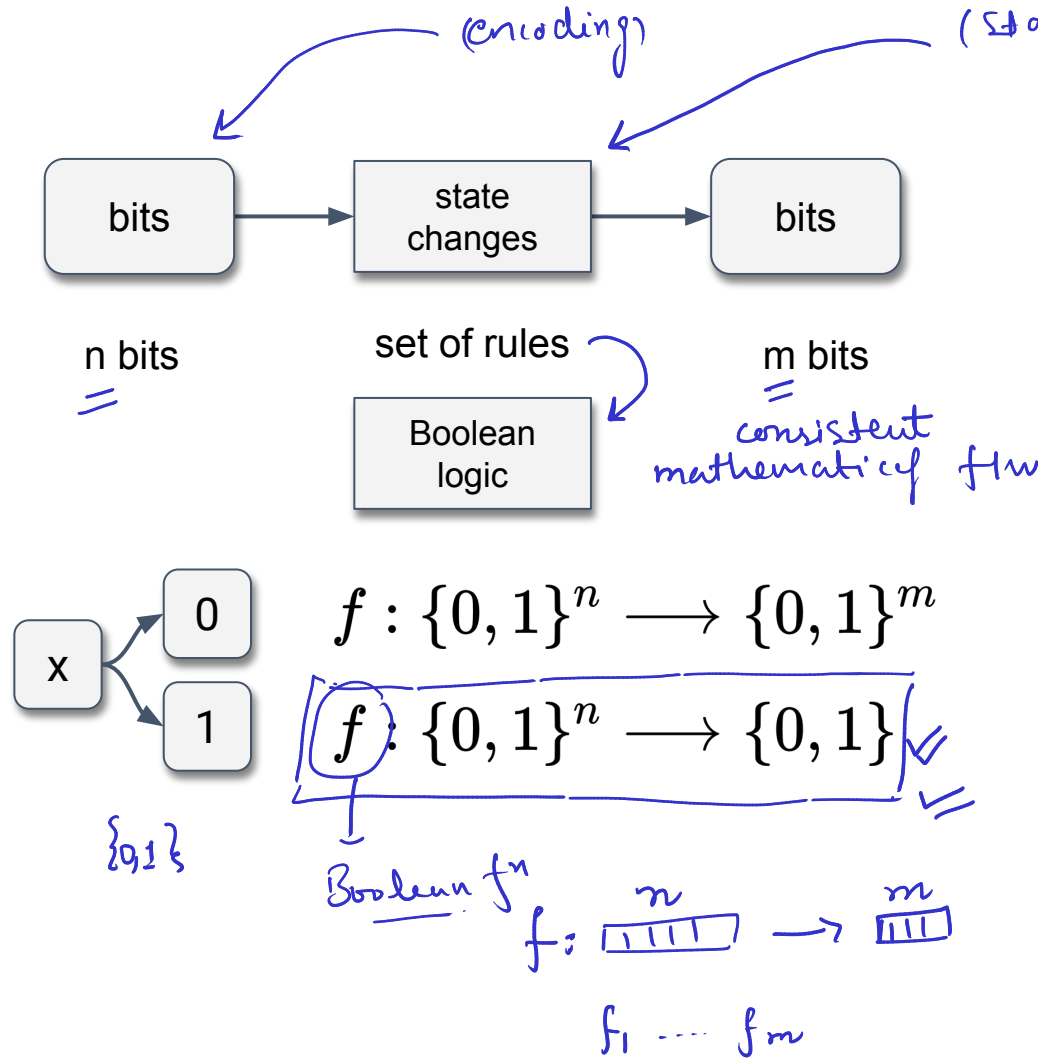


Classical information and computation



Bits, Functions, Circuits, Algorithms

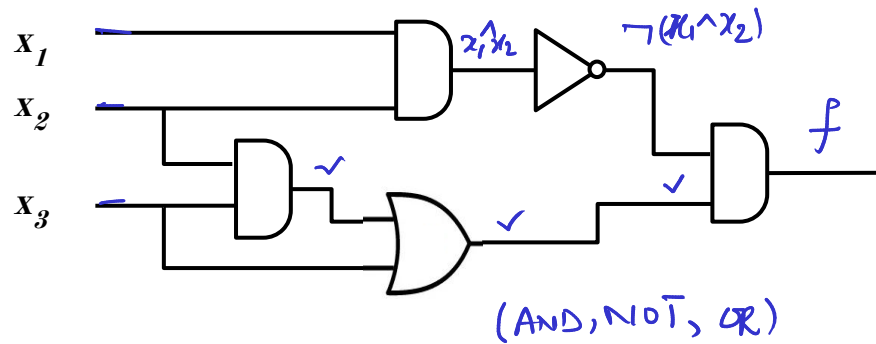
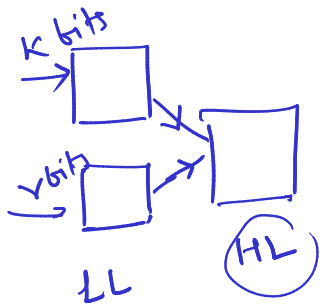
- A binary digit (bit) is a two-level system
 - a bit can assume two states - 0 and 1
 - fundamental unit of classical information
- Operations on bits are carried out
 - Boolean logic formalism
 - Logic circuits describe a boolean function
- On the physical (device) level
 - implemented using transistors as switches
 - operations are gated operations $\rightarrow \begin{matrix} \rightarrow 1 \\ \rightarrow 0 \end{matrix}$

Logic circuits as computational model

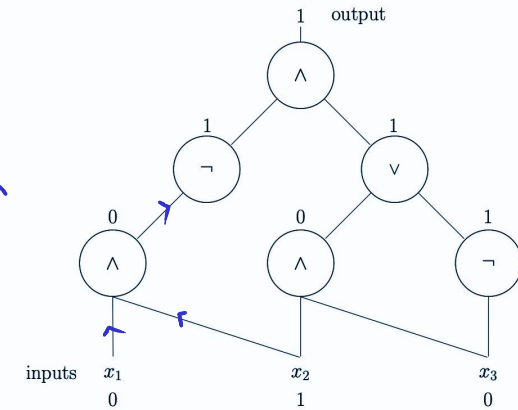
3 bits (3 binary variables)

formal

$$f(x_1, x_2, x_3) = (\neg(x_1 \wedge x_2)) \wedge ((x_2 \wedge x_3) \vee (\neg x_3)) \quad \checkmark$$



Logic circuit



Computation graph

- Boolean functions are implemented as logic circuits
 - is there a **universal set of logic gates**?
 - {NOT, AND} is a universal set for implementing any Boolean function
- Complex computations can be composed of simpler computations
 - modularity and levels of abstraction
 - fundamentally, all operations are bit-level operations

Exercise:

1. Quantum gates and quantum circuits?
2. *Universal set of quantum gates?*

